

Matthew Louis Davis

321 N Johnson St Apt 8, Iowa City, IA 52245 | 847-440-6841 | matthewlouisdavis@gmail.com

EDUCATION

Ph.D. Biostatistics

(expected) May 2023

University of Iowa, Iowa City, IA

Thompson Sampling with Smoothing Splines for Hyperparameter Optimization: A Tuning Algorithm That Does Not Require Tuning: Advised by Brian J. Smith

M.S. Biostatistics

May 2020

University of Iowa, Iowa City, IA

B.S. Molecular Cellular Biology, Minor in Mathematics

May 2016

University of Puget Sound, Tacoma, WA

RELEVANT COURSEWORK

Biostatistical Computing: Programming with R	Survey Design and Analysis	Biostatistical Methods I
Biostatistical Computing: Programming with SAS	Survival Data Analysis	Mathematical Statistics I
Advanced Biostatistical Computing	Longitudinal Data Analysis	Mathematical Statistics II
Machine Learning for Biomedical Data	Statistical Methods in Clinical Trials	Statistical Inference II
Big Data Analysis with Python	Advanced Clinical Trials	Theory Linear/Generalized Linear Models
Bayesian Methods and Design	Biostatistical Methods Categorical Data	Likelihood Theory and Extensions

SOFTWARE SKILLS

Languages: R, SAS, Python, Julia, SQL, LaTeX

Computing Environments: Windows, Argon High Performance Computing

Software: Google BigQuery, PowerBI, Microsoft Office

WORK EXPERIENCE

Course Instructor (June 2022 – Present)

BIOS 4120: Introduction to Biostatistics

University of Iowa, College of Public Health, IA

- Served as head instructor of introductory biostatistics course at a large public university
- Lectured online over Zoom four times a week for 95 minutes for 6 weeks (summer), and twice a week for 80 minutes for 16 weeks (fall)
- Designed presentations, assignments, projects, and exams using LaTeX
- Educated non-statistical audiences on statistical theory (e.g. Regression, Bayes Rule, Hypothesis Testing)
- Supported struggling students with 1-on-1 assistance outside of classroom hours
- Received 89/90 on peer evaluation, and inspired several students to pursue biostatistics as a career

Research Assistantships

University of Iowa, Office of Teaching and Learning Technologies, IA (May 2019 – May 2022)

- Analyzed undergraduate student data to help faculty learn about student behaviors and professors to develop more effective teaching strategies

- Constructed PowerBI dashboards to assist faculty in understanding the behaviors of their students, allowing for crucial insights of online behaviors during the COVID-19 pandemic
- Communicated important findings to faculty and professors, such as which course materials students accessed, and which student behaviors were most associated with academic performance
- Used SQL and API queries to gather data from online sources and clean for formal analysis
- Applied statistical and machine learning methods to generate insights from big data sets

National Resource Center, Iowa School of Social Work, IA (March 2019 – August 2019)

- Collaborated as the sole statistician with a diverse team of non-statisticians including social workers and other researchers
- Compared the effectiveness of social work interventions using contingency table analyses
- Designed questionnaires for understanding the sentiments of participants pre-vs-post interventions
- Wrote SAS macros to automate the simultaneous generation of tabular data, saving time and effort for social workers who would otherwise construct each table 1-by-1 manually

Teaching Assistantships

BIOS 5310: Research and Data Management

University of Iowa, College of Public Health, IA (August 2018– May 2019)

- Graded homework and coursework for an introductory SAS course at a large public university
- Helped students lacking a formal programming background learn how to code

BIOS 4120: Introduction to Biostatistics

University of Iowa, College of Public Health, IA (August 2018– May 2019)

- Graded homework and coursework for an introductory biostatistics course at a large public university
- Assisted students of diverse academic backgrounds understand and use basic statistical concepts

Emergency Department Scribe (June 2016 – November 2016)

PhysAssist, Tacoma, WA

- Responsible for formally documenting the interaction between a patient and a physician
- Gained shadowing experience with doctors and formed relationships with medical professionals
- Documented in detail with appropriate medical terminology: Laboratory findings, Urine and blood analysis, EKG results, Ultrasound procedures, Laceration repair procedures, CT scans and MRI interpretations, X-Ray findings, etc.
- Worked 8-12 hour shifts in an emergency department setting with over 25 different providers in 4 separate ED locations around the greater Seattle area, in varying economic and social environments
- Assessed with 76/80 for overall performance and professionalism after 450 hours of service

ADDITIONAL EXPERIENCE

METHODOLOGICAL RESEARCH

Matthew L Davis, Yuan Huang, Kai Wang, Rank normalization empowers a *t*-test for microbiome differential abundance analysis while controlling for false discoveries, *Briefings in Bioinformatics*, Volume 22, Issue 5, September 2021, bbab059, <https://doi.org/10.1093/bib/bbab059>

INTERNSHIP

Northwestern University, Feinberg School of Medicine, Chicago IL (May 2017 – July 2017)

- Prepared and organized data specifications for the Chicago Multicenter AIDS Cohort Study
- Produced summary statistics in R, data cleaning in excel, and Circos in Perl to visualize the spread of

C. difficile between hospital departments

- Performed ANOVA relating cohort demographics to clinically relevant outcomes
- Recorded, entered, and verified the validity of raw, level-I data in an online biomedical database

ACTIVITIES

Biostatistics Student Organization (August 2018 – Present)

University of Iowa, Iowa City, IA

- Represented the biostatistics department at the Graduate Student Senate from fall 2018 to spring 2019
- Served as volunteer and activities coordinator from fall 2021 to spring 2022
- Mentored incoming students on succeeding in graduate school

University of Puget Sound Swim Team (September 2012– February 2016)

University of Puget Sound, Tacoma, WA

- Four-year varsity swimmer
- Team Captain 2015-2016

FIMRC Volunteer (October 2015 -November 2015)

FIMRC Huancayo and La Merced, Peru

- Volunteered for medical and education services for 4 weeks in two small Peruvian cities
- Worked with a small multi-national team of 4 different nurses and volunteers to address various needs of local communities
- Recorded vitals such as heart rate, blood pressure, blood glucose levels, respiration rate, arterial blood pressure, SpO₂ saturation, height, and weight (up to 80 patients a day) under physician supervision.
- Developed a proficient and extensive vocabulary of Spanish medical terminology
- Designed 3 educational programs for children in native communities and orphanages for health, hygiene, and recycling while crossing 2 language barriers (Quechua and Spanish)